

ARMAVIR, Russian Federation

WMO: 370310

Lat: 44.983N

Lon: 41.117E

Elev: 522

StdP: 14.42

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.4	11.7	0.1	5.6	6.6	5.6	7.5	13.3	23.4	34.4	21.7	33.4	4.4	80	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	94.8	69.7	91.7	68.9	88.2	68.2	73.7	86.7	72.0	84.7	70.7	82.9	6.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.8	111.6	80.1	68.2	105.5	77.3	66.8	100.4	76.1	37.7	86.8	36.2	84.2	34.9	83.2	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.7	16.8	14.0	DB	-2.2	99.0	7.6	3.4	-7.6	101.4	-12.0	103.4	-16.3	105.3	-21.8	107.8	(4)
(5)			WB	-2.0	75.9	7.8	2.5	-7.6	77.7	-12.2	79.2	-16.6	80.6	-22.2	82.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.5	31.1	33.9	42.5	53.2	62.5	70.0	75.4	74.6	65.2	53.8	42.6	35.6		
	DBStd	17.23	9.55	10.14	7.61	7.41	6.88	5.40	4.90	5.61	6.86	8.38	8.49	8.41		
	HDD50	2089	587	451	252	48	3	0	0	0	2	53	245	448		
	HDD65	5147	1052	870	697	357	130	16	1	3	80	357	671	913		
	CDD50	3366	0	2	20	145	391	601	788	763	458	173	24	1		
	CDD65	947	0	0	0	4	53	167	324	301	86	12	0	0		
	CDH74	9501	0	0	3	62	531	1461	3370	3215	756	102	1	0		
	CDH80	3867	0	0	0	9	128	499	1497	1497	212	25	0	0		
(14) Wind	WSAvg	5.5	6.0	6.5	7.3	6.2	5.0	4.7	4.5	4.6	5.0	5.1	5.9	6.0		
(15) Precipitation	PrecAvg	22.5	1.3	1.2	1.5	2.1	2.6	2.8	2.2	2.2	1.7	1.6	1.7	1.6		
	PrecMax	32.9	2.8	3.2	2.9	6.1	4.8	8.1	5.5	5.4	3.4	3.5	3.5	4.3		
	PrecMin	15.2	0.3	0.0	0.0	0.2	0.2	0.1	0.4	0.4	0.7	0.2	0.2	0.3		
	PrecStd	4.1	0.6	0.8	0.7	1.4	1.1	1.9	1.3	1.6	0.8	1.0	0.9	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	61.9	70.2	80.6	87.5	93.2	98.0	99.2	89.9	82.8	68.5	56.7		
		MCWB	44.7	48.8	52.8	60.1	67.0	69.8	71.0	68.4	68.6	63.0	53.8	47.7		
	2%	DB	47.3	54.9	64.3	74.8	83.8	89.2	94.7	95.6	85.6	76.2	62.2	51.4		
		MCWB	42.1	45.9	49.9	58.7	65.4	68.9	70.1	69.5	65.2	61.3	52.0	45.1		
	5%	DB	44.1	49.9	59.0	70.6	80.0	85.7	91.7	92.1	82.0	70.8	57.6	48.1		
		MCWB	40.2	42.8	48.3	56.4	63.9	67.8	69.0	68.7	64.4	59.0	50.1	43.8		
	10%	DB	41.2	45.6	54.2	66.4	76.4	82.0	88.1	88.3	78.3	65.9	53.5	45.2		
		MCWB	37.8	40.5	46.1	54.5	62.6	66.7	68.6	68.0	62.9	57.0	48.1	41.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.7	49.7	54.0	63.0	69.9	74.1	76.9	75.3	71.7	64.5	55.8	48.6		
		MCDB	50.7	59.2	66.6	75.2	83.9	86.2	91.6	89.2	85.7	78.1	63.5	54.9		
	2%	WB	42.7	46.4	51.3	59.8	67.3	71.7	74.2	73.4	68.3	62.3	53.2	46.0		
		MCDB	46.6	54.4	61.8	71.8	79.8	84.0	86.8	87.2	79.8	73.5	60.7	50.5		
	5%	WB	40.3	43.4	49.1	57.5	65.5	70.0	72.3	71.8	66.5	60.1	50.8	44.1		
		MCDB	43.9	49.3	57.5	67.8	76.7	81.5	84.4	85.0	77.1	69.1	57.2	47.8		
	10%	WB	38.1	40.8	46.6	55.6	63.7	68.3	70.8	70.3	64.8	58.0	48.2	41.9		
		MCDB	41.1	45.3	53.5	64.6	73.4	78.6	82.7	82.6	74.8	65.0	52.8	45.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.0	14.4	15.7	18.5	18.9	19.1	21.4	22.5	20.4	17.1	14.7	11.6		
		MCDBR	15.2	20.9	25.6	25.5	25.6	25.3	26.8	28.8	26.3	26.1	22.6	15.9		
		MCWBR	10.6	12.6	13.4	12.0	11.0	9.4	9.0	9.1	10.1	12.1	12.9	10.9		
	5% WB	MCDBR	14.4	19.6	22.6	21.9	22.3	21.3	21.8	22.8	21.9	22.4	20.7	14.8		
		MCWBR	10.4	12.3	12.7	11.2	10.6	9.3	8.9	8.9	9.7	10.8	12.2	10.4		
(40) Clear-Sky Solar Irradiance	taub		0.337	0.370	0.417	0.463	0.427	0.434	0.460	0.457	0.421	0.388	0.351	0.338		
	taud		2.297	2.206	2.121	2.036	2.227	2.244	2.187	2.196	2.268	2.319	2.364	2.315		
	Ebn at Noon		239	252	255	253	268	266	257	252	252	243	233	225		
	Edh at Noon		26	35	43	51	43	43	45	43	37	30	24	23		
(44) All-Sky Solar Radiation	RadAvg		445	727	997	1472	1807	1977	2016	1812	1342	855	516	370		
	RadStd		60	101	89	150	199	121	98	106	146	105	59	40		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (9 neighbors)		+0.86	N/A	N/A	N/A	N/A	N/A	N/A	-220	+212	N/A	

Nomenclature: See separate page